

Model fitting: Maximum Likelihood estimation

Once we have examined the sample ACF and PACF plots and have decided on a few candidate ARIMA(p,d,q) models, we fit them using maximum likelihood estimation (MLE). **We do NOT concern ourselves with estimating the parameters in this course by hand i.e. we let Rstudio do its job!!!** However, some basic understanding of MLE is important:

$$\mu = \dots \quad \sigma^2 = \dots$$

When R estimates the ARIMA model, it uses maximum likelihood estimation (MLE). This technique finds the values of the parameters (i.e. model coefficients) which maximize the probability of obtaining the data that we have observed i.e. MLE adjusts the estimated coefficient values until the probability that the time series data we have observed comes from the specified ARIMA(p,d,q) model is maximized.

$$\log(\prod_{i=1}^n f(x_i; \theta)) = \sum_{i=1}^n \log(f(x_i; \theta)) \quad \leftarrow \prod_{i=1}^n f(x_i; \theta) = f(x_1) \cdot f(x_2) \cdot \dots \cdot f(x_n)$$

In practice, R will report the value of the log likelihood of the data; that is, the logarithm of the probability of the observed data coming from the estimated model. For given values of p, d and q, R will try to maximize the log likelihood when finding the coefficient estimates.

Given appropriate model output, you should be able to explain what the log-likelihood, $\log(L)$, is. You should also need to be able to state the fitted model and calculate its properties (e.g. mean, variance and autocorrelation at a certain lag, etc.)



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Model selection: Using AIC

Once we have examined the sample ACF and PACF plots and have decided and fitted a few ARIMA(p,d,q) models to the time series, how are we to decide which model is the best one to fit to the data?

ANSWER: Use information criteria! Essentially, information criteria trade off the goodness-of-fit of the model with the complexity (i.e. order of the model). Lower order models that fit the data better than other models are preferred. The most commonly used information criteria are the Bayes Information Criterion (BIC), the Akaike Information Criterion (AIC) and the corrected Akaike Information Criterion (AICc). **We will use AIC in this course.**

The Akaike information criterion is a measure of the relative quality of a statistical model for a given set of data. That is, given a collection of models for the data, AIC estimates the quality of each model, relative to each of the other models. Hence, AIC provides a means for model selection:

$$\text{AIC} = -2 \log(L) + 2(p + q + k + 1)$$

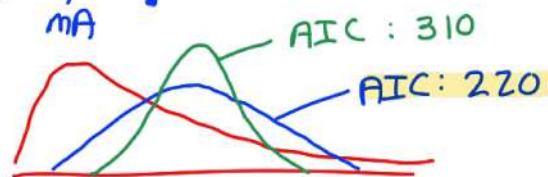
↓ ↗ penalty
AR MA

Where:

L is the likelihood of the data

$k = 1$ if $c \neq 0$ and $k = 0$ if $c = 0$

p, d and q refer to the number of autoregressive terms, order of differencing and number of moving average terms



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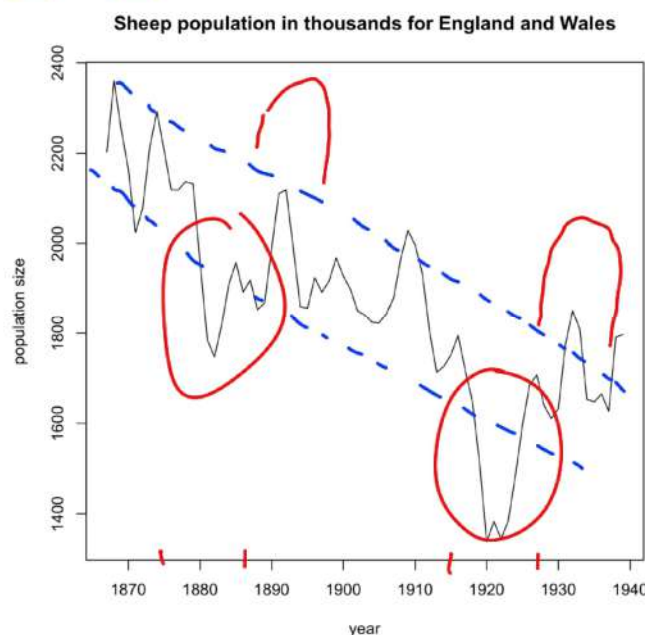
Model selection: Using AIC

- The first term of the expression is a measure of how well the model fits the data.
- The variance of the forecasts of a model (due to estimation error) increases with the overall number of terms in the model. This implies that you don't want a forecasting model with too many terms.
- But how many is "too many"? The second term represents a "penalty" term for the order (complexity) of the fitted model i.e. this term is a "penalty" for using more terms, which increases the forecast variance.
- **Minimizing AIC trades off the goodness-of-fit with the number of fitted terms to determine a "best" model for your forecast.** This trade-off is a well-known phenomenon in statistics that is referred to as the bias-variance trade-off.
- A smaller value for AIC for a model (when compared to other models) implies a model with fewer parameters, a better fit OR both (i.e. fewer parameters and better fit).
- **NB: AIC does not provide a test of a model in the sense of testing a null hypothesis; i.e. AIC can tell nothing about the quality of the model in an absolute sense. If all the candidate models fit poorly, AIC will not give any warning of that.**
- **To use the AIC to select a model in practice**, one computes the AIC for a variety of model orders (e.g. for an ARIMA(2,1,1) model, the order = 3) and then select the model that has the **minimum AIC value**.

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ARIMA time series approach --- Example

The time series of the annual sheep population (in thousands) of England and Wales is plotted for the years 1867 to 1939 in the diagram below:

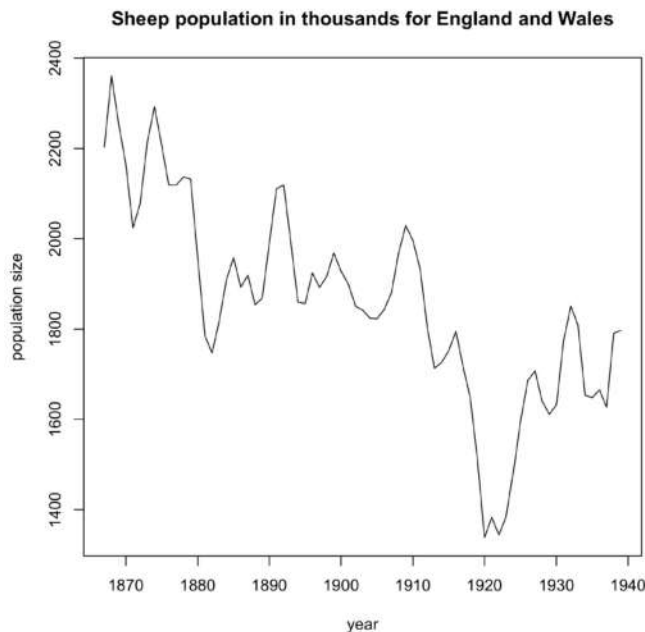


stationary?
— trend: mean,
not constant!
— variation:
relatively
constant

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ARIMA time series approach --- Example

The time series of the annual sheep population (in thousands) of England and Wales is plotted for the years 1867 to 1939 in the diagram below:

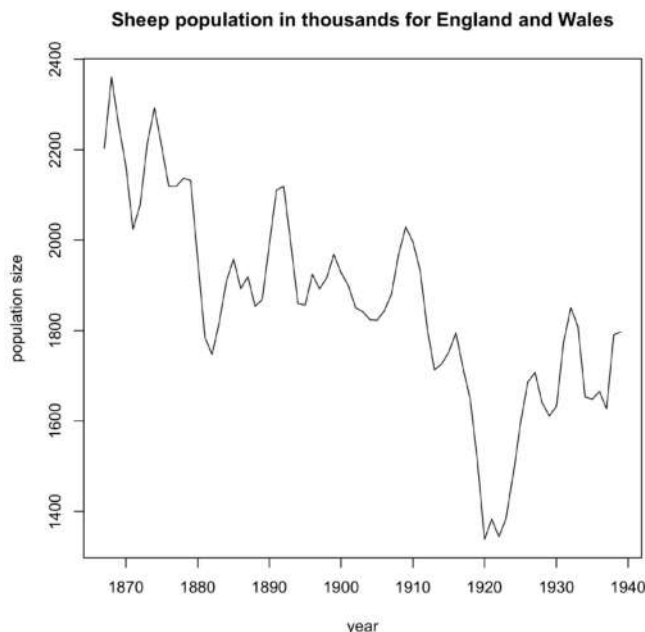


a) Is the time series stationary? Justify your answer.

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ARIMA time series approach --- Example

The time series of the annual sheep population (in thousands) of England and Wales is plotted for the years 1867 to 1939 in the diagram below:



a) Is the time series stationary? Justify your answer.

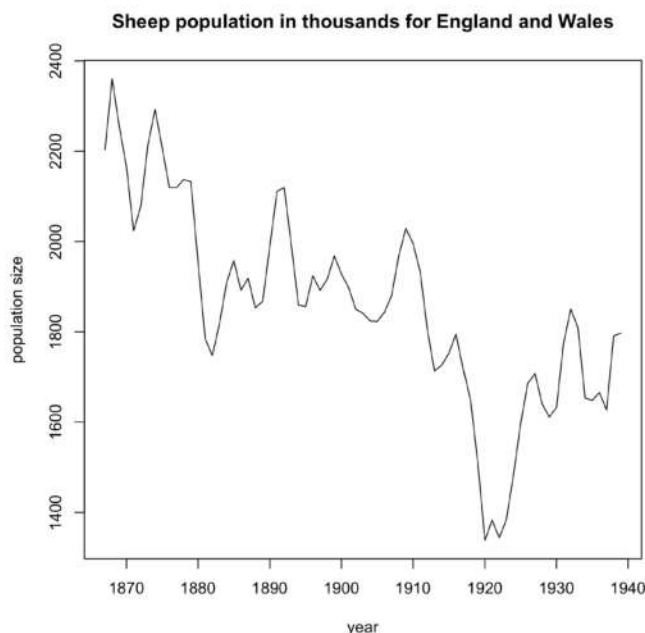
No. There is a clear downward **approx** linear(potentially non-linear*) trend in the series. Hence, the series is not stationary in mean.

*For the purposes of this example, we go with the trend being linear and decreasing.

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ARIMA time series approach --- Example

The time series of the annual sheep population (in thousands) of England and Wales is plotted for the years 1867 to 1939 in the diagram below:



b) What transformations should you apply to this series to result in a stationary series?

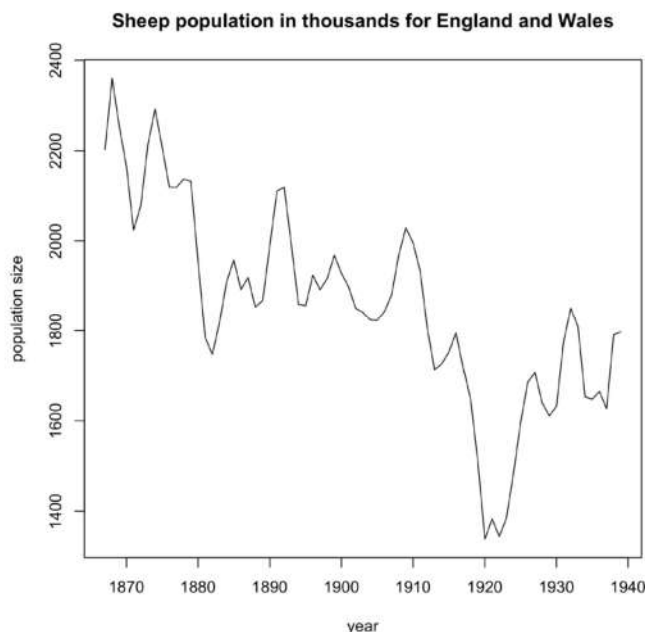
- BoxCox (good idea)
- diff @ lag 1

Worst

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ARIMA time series approach --- Example

The time series of the annual sheep population (in thousands) of England and Wales is plotted for the years 1867 to 1939 in the diagram below:



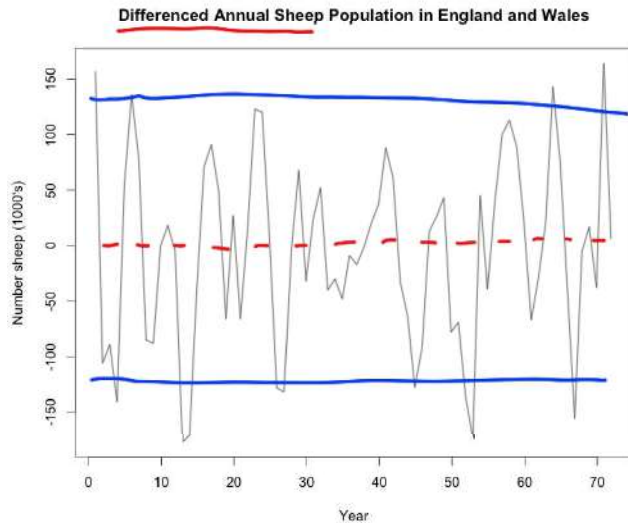
b) What transformations should you apply to this series to result in a stationary series?

The variance appears relatively constant, so there are no Box-Cox transformations needed in that regard. To remove the trend, you should apply a first-order of differencing at lag 1 i.e.

$$Y_t = X_t - X_{t-1}, \quad t = 2, 3, \dots, 73$$

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ARIMA time series approach --- Example

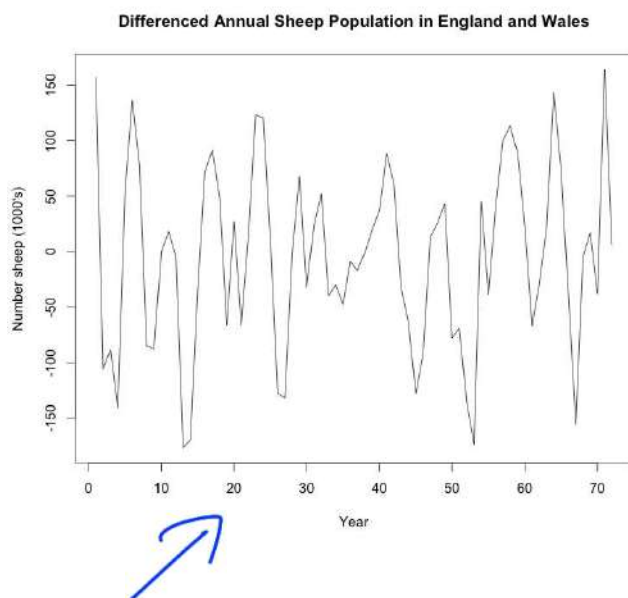


c) A first-order difference at lag 1 was performed, and the differenced series plotted as shown in the figure to the left. Comment on the stationarity of the difference series. Are any more transformations required?

No! Looks Stationary.

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ARIMA time series approach --- Example



c) A first-order difference at lag 1 was performed, and the differenced series plotted as shown in the figure to the left. Comment on the stationarity of the difference series. Are any more transformations required?

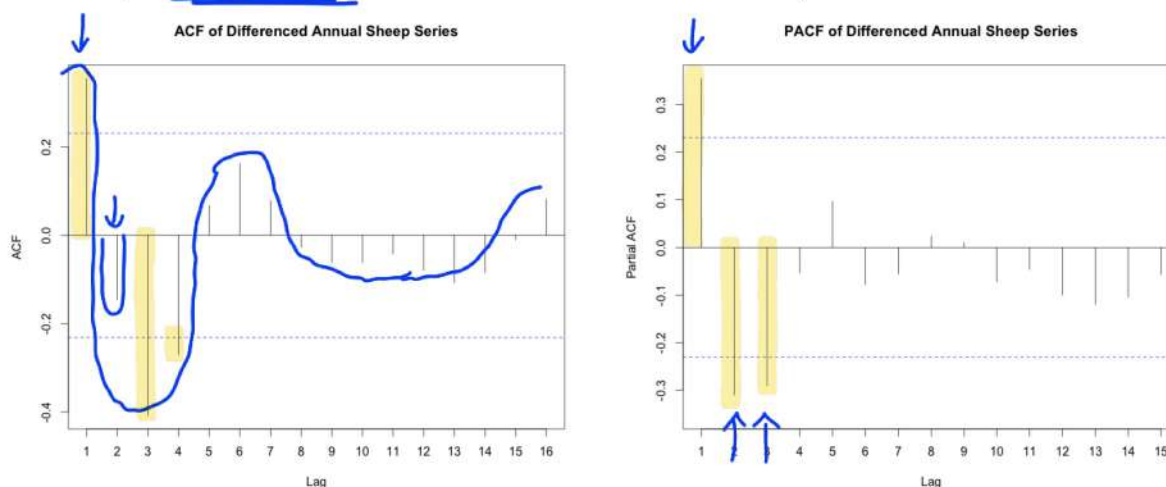
The differenced series appears stationary – it has a relatively constant mean and variance.

Therefore, no further transformations are necessary.

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ARIMA time series approach --- Example

The sample ACF and PACF plots of the transformed series are plotted below:



d) Suggest an appropriate type of model to fit to this residual series, and hence an appropriate model to fit to the original data. Can you suggest what order of model to fit? Provide reasons for your answer **ARIMA(3,1,0)**

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ARIMA time series approach --- Example

d) Suggest an appropriate type of model to fit to this transformed series, and hence an appropriate model to fit to the original data. Can you suggest what order of model to fit? Provide reasons for your answers.

Both the ACF and PACF shut off after lag 4 and 3 respectively. This behaviour is representative of a combined ARIMA(p,d,q) process, Hence we should fit a few low-order ARIMA models and compare them with the AIC.

If we were to fit an AR(p) model, an AR(3) model should be fit. Also, it must be noted that although the partial autocorrelations at lags 1, 2, 3 are significantly different from 0, they are only moderate in strength (just smaller than -0.3).

Since a first order difference was applied to result in a stationary series, we are fitting a non-seasonal **ARIMA(3,1,0)** model to the original data. $\rightarrow \mu = \frac{c}{1-\phi_1-\phi_2-\phi_3}$

Let us suppose one of the ARIMA models fit was a non-seasonal ARIMA(3,1,0). The output from fitting the model is included below:

<u>ARIMA(3,1,0)</u>	ϕ_1	ϕ_2	ϕ_3	μ	
Coefficients: ar1	ar2	ar3	mean		intercept
0.4134	-0.2045	-0.3115	-5.8707	$\neq c$	
s.e. 0.1200	0.1367	0.1250	7.5076		

sigma² estimated as 4809: log likelihood=-401.6
AIC=813.2 AICc=814.11 BIC=824.58

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ARIMA time series approach --- Example

ARIMA(3,1,0)

Coefficients:	ar1	ar2	ar3	mean	
	0.4134	-0.2045	-0.3115	-5.8707	= μ
s.e.	0.1200	0.1367	0.1250	7.5076	

sigma^2 estimated as 4809: log likelihood=-401.6

AIC=813.2 AICc=814.11 BIC=824.58

$N(0, \sigma^2)$

e) State the fitted model. What is the value of the constant term in the model?

$$X_t = C + \phi_1 X_{t-1} + \phi_2 X_{t-2} + \phi_3 X_{t-3} + \varepsilon_t$$

$$= C + 0.4134 X_{t-1} - 0.2045 X_{t-2} - 0.3115 X_{t-3} + \varepsilon_t$$

$$C = \mu(1 - \phi_1 - \phi_2 - \phi_3)$$

$$= -5.8707(1 - 0.4134 + 0.2045 + 0.3115) = -6.4730$$

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ARIMA time series approach --- Example

ARIMA(3,1,0) with zero mean

Coefficients:	ar1	ar2	ar3	mean
	0.4134	-0.2045	-0.3115	-5.8707
s.e.	0.1200	0.1367	0.1250	7.5076

sigma^2 estimated as 4809: log likelihood=-401.6

AIC=813.2 AICc=814.11 BIC=824.58

e) State the fitted model. What is the value of the constant term in the model?

$$X_t = c + 0.4134X_{t-1} - 0.2045X_{t-2} - 0.3115X_{t-3} + Z_t$$

The estimated mean of the AR(3) series is given by -5.8707. Therefore, the constant is calculated as:

$$c = -5.8707(1 - 0.4134 + 0.2045 + 0.3115) = -6.4730$$

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ARIMA time series approach --- Example

ARIMA(3,1,0) with zero mean

Coefficients:	ar1	ar2	ar3	mean
	0.4134	-0.2045	-0.3115	-5.8707
s.e.	0.1200	0.1367	0.1250	7.5076

σ^2 estimated as 4809: log likelihood=-401.6

AIC=813.2 AICc=814.11 BIC=824.58

f) Interpret the value of the AIC

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ARIMA time series approach --- Example

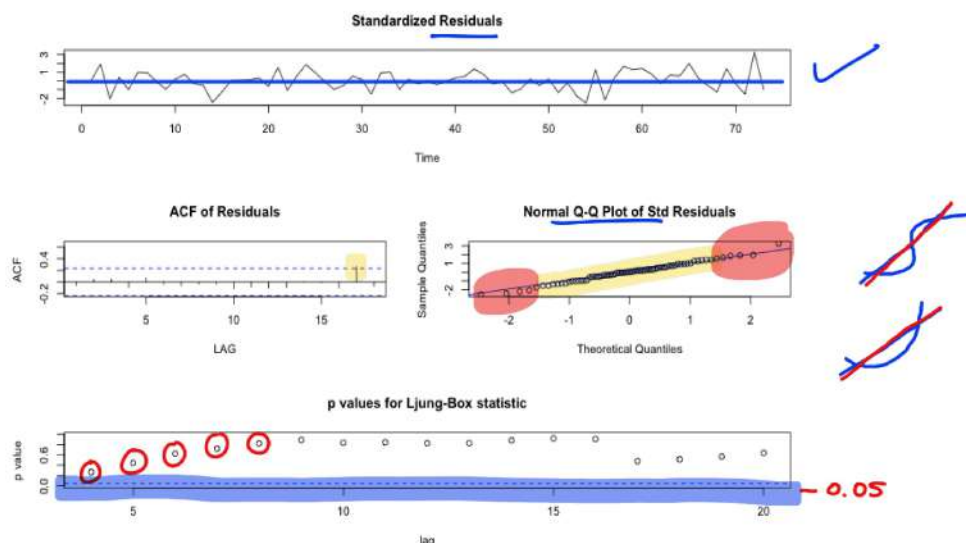
f) Interpret the values of AIC

The AIC value does not have a direct interpretation in this context as we fitted only 1 model, and thus were not comparing fitted models with one another.

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ARIMA time series approach --- Example

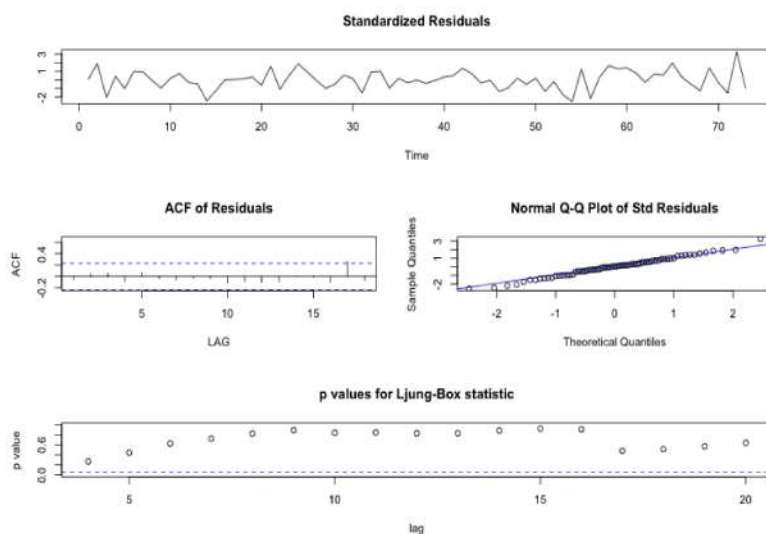
Residual diagnostics were carried out on the fitted ARIMA(3,1,0) model, as displayed in the figure below:



h) Is the ARIMA(3,1,0) an adequate model to forecast values of the Sheep population data? Justify your answer by referring to *each* element of the output

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ARIMA time series approach --- Example



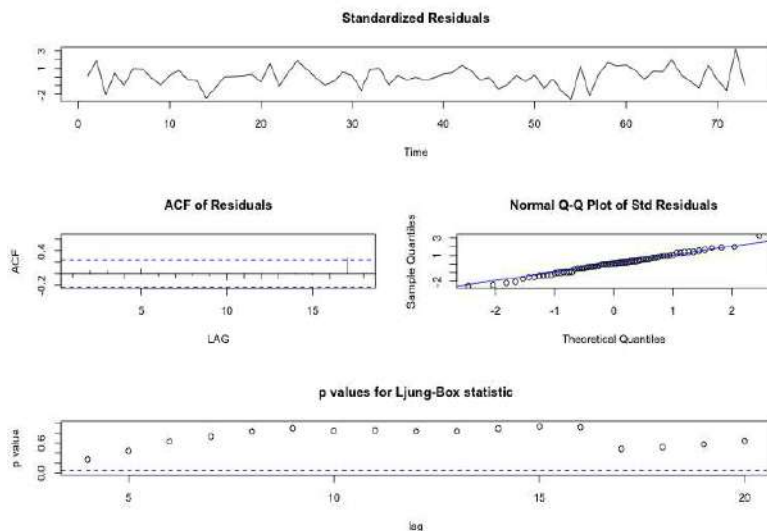
h) Is the ARIMA(3,1,0) an adequate fit to the Sheep population data? Justify your answer by referring to *each* element of the output

YES. The plot of the standardised residuals does not show a pattern, and they appear to have a mean of 0 and relatively constant variance.

The ACF of the residuals contains very small values apart from the one at lag 17. It is possible that this is significant due to chance or sampling error (there is only 1 out of 20 ACF values that are significantly different from 0).

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ARIMA time series approach --- Example



The normal Q-Q plot does not indicate any serious deviations from normality.

The p-values for the Ljung—Box test are all > 0.05 , indicating that we would not reject the null hypothesis that the residuals are independent at any of the first 20 lags.

Hence, the ARIMA(3,1,0) is an adequate fit to the original data.

Note that you should be comfortable interpreting both residual diagnostic output like this and the output from the `checkresiduals()` function.

ALWAYS comment on each part of the output when assessing the residuals of a fitted model.

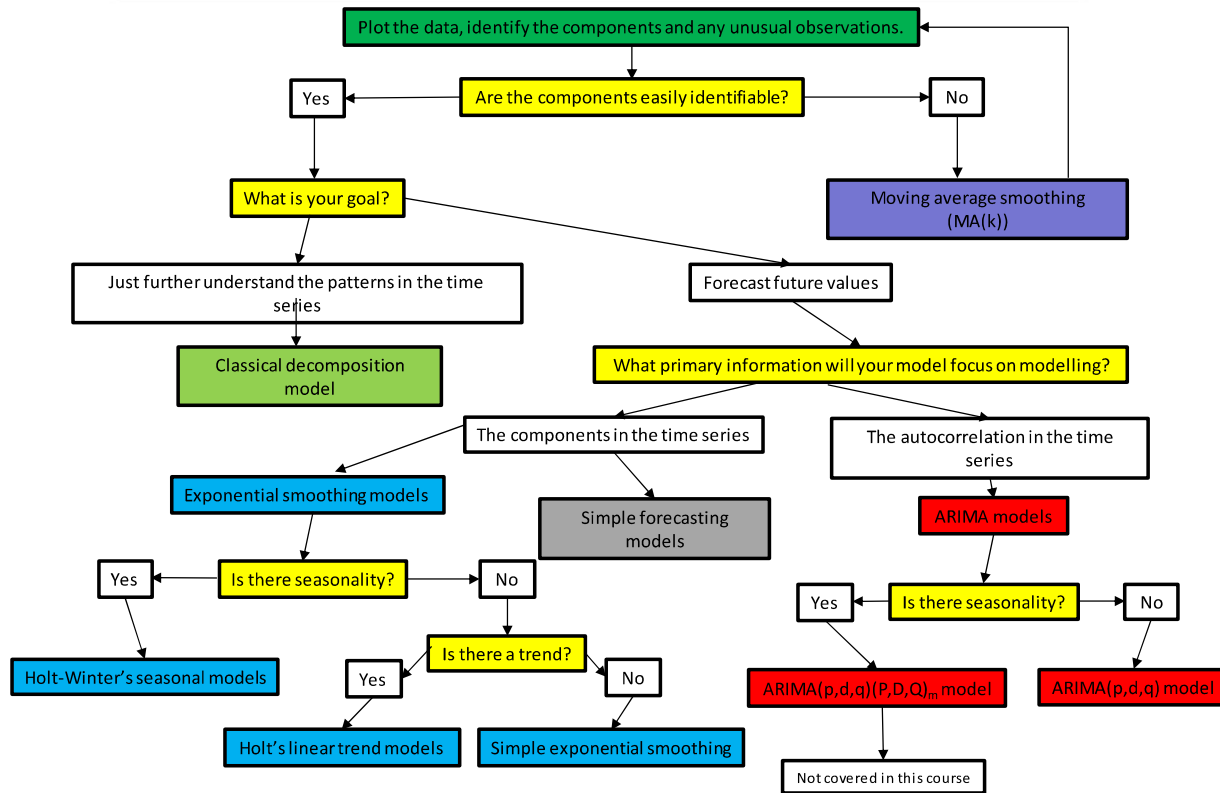
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Overall time series process examples

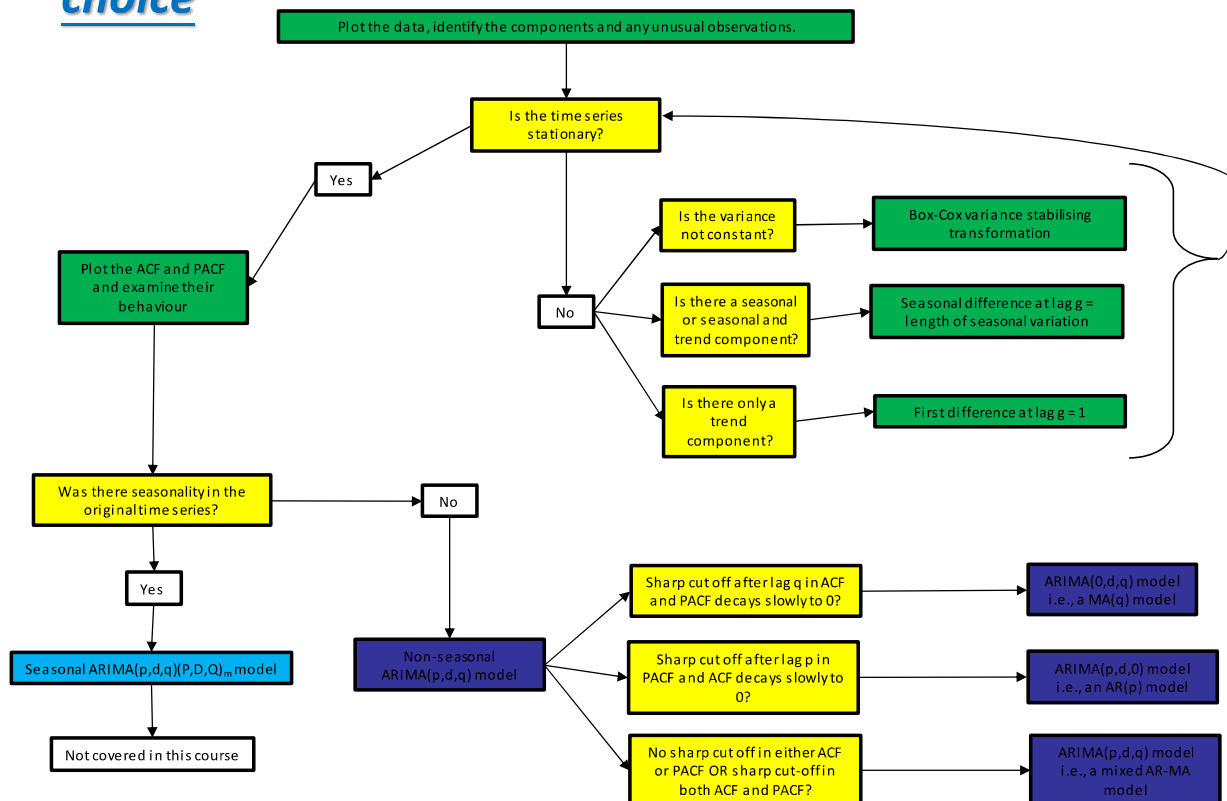
Roadmaps & Conducting a time series analysis from start to finish

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General time series approach --- Model choice



General time series approach --- ARIMA model choice



General time series approach --- Exponential Smoothing model choice (with simple forecasting methods)

